



Date: 09 / 01 / 2026

Lab Practical #6:

Implementation of Hill Cipher Technique.

1. 2x2 key matrix
2. 3x3 key matrix

Program:

```
#include <stdio.h>
#include <string.h>

int main() {
    int key[2][2] = {{3, 3}, {2, 5}};
    char plaintext[100];
    int i;

    printf("Enter Plaintext (uppercase): ");
    scanf("%s", plaintext);

    // If length is odd, add X
    if (strlen(plaintext) % 2 != 0) {
        strcat(plaintext, "X");
    }

    printf("Ciphertext: ");

    for (i = 0; i < strlen(plaintext); i += 2) {
        int p1 = plaintext[i] - 'A';
        int p2 = plaintext[i + 1] - 'A';

        int c1 = (key[0][0]*p1 + key[0][1]*p2) % 26;
        int c2 = (key[1][0]*p1 + key[1][1]*p2) % 26;

        printf("%c%c", c1 + 'A', c2 + 'A');
    }

    return 0;
}
```

Output:

```
D:\CSE B-Tech\Semester-6\INS\Lab-6>gcc two_matrix.c
D:\CSE B-Tech\Semester-6\INS\Lab-6>a
Enter Plaintext (uppercase): OK
Ciphertext: UA
```

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Program:

```
#include <stdio.h>
#include <string.h>

int main() {
    int key[3][3] = {
        {6, 24, 1},
        {13, 16, 10},
        {20, 17, 15}
    };

    char plaintext[100];
    int i;

    printf("Enter Plaintext (uppercase): ");
    scanf("%s", plaintext);

    // Padding with X if needed
    while (strlen(plaintext) % 3 != 0) {
        strcat(plaintext, "X");
    }

    printf("Ciphertext: ");

    for (i = 0; i < strlen(plaintext); i += 3) {
        int p[3], c[3], j;

        for (j = 0; j < 3; j++) {
            p[j] = plaintext[i + j] - 'A';
        }

        for (j = 0; j < 3; j++) {
            c[j] = (key[j][0]*p[0] + key[j][1]*p[1] + key[j][2]*p[2]) % 26;
            printf("%c", c[j] + 'A');
        }
    }

    return 0;
}
```



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Output:

```
D:\CSE B-Tech\Semester-6\INS\Lab-6>gcc three_matrix.c  
  
D:\CSE B-Tech\Semester-6\INS\Lab-6>a  
Enter Plaintext (uppercase): ASK  
Ciphertext: AY0
```